

Mid-Term Examination for Quantum Mechanism, Semester of 2025-2026(2)

司徒和丞相 2024001

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Problem 1

A particle of mass m has the wave function:

$$\Psi(x, t) = Ae^{(-a[\frac{mx^2}{\hbar} + it])}$$

where A and a are positive real constants.

- (a) Find A .
- (b) For what potential energy function $V(x)$ is this a solution to the Schrödinger equation?
- (c) Calculate the expectation values of x , x^2 , p and p^2 .
- (d) Find σ_x and σ_p . Is their product consistent with the uncertainty principle?

Solution.

(a) Normalize: Find A

The normalization condition is $\int_{-\infty}^{\infty} |\Psi|^2 dx = 1$.

The modulus squared is (the e^{-iat} and e^{+iat} cancel):

$$|\Psi|^2 = A^2 \exp\left(-2am\frac{x^2}{\hbar}\right)$$

Let $\alpha \stackrel{\text{def}}{=} 2a\frac{m}{\hbar}$. Using the Gaussian integral:

$$A^2 \int_{-\infty}^{\infty} e^{-\alpha x^2} dx = A^2 \sqrt{\frac{\pi}{\alpha}} = A^2 \sqrt{\frac{\pi\hbar}{2am}} = 1$$

$$\boxed{A = \left(\frac{2am}{\pi\hbar}\right)^{\frac{1}{4}}}$$

(b) Find $V(x)$ from the Schrödinger Equation

The time-dependent Schrödinger equation:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V(x)\Psi$$

Left side — time derivative:

$$\frac{\partial \Psi}{\partial t} = -ia\Psi \Rightarrow i\hbar \frac{\partial \Psi}{\partial t} = \hbar a\Psi$$

Right side — second spatial derivative:

$$\frac{\partial \Psi}{\partial x} = -\frac{2amx}{\hbar}\Psi$$

$$\frac{\partial^2 \Psi}{\partial x^2} = \left(-\frac{2am}{\hbar} + \frac{4a^2m^2x^2}{\hbar^2}\right)\Psi$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} = (\hbar a - 2a^2 m x^2) \Psi$$

Substituting into the Schrödinger equation:

$$\hbar a = \hbar a - 2a^2 m x^2 + V(x)$$

$$\boxed{V(x) = 2 a^2 m x^2}$$

This is a *harmonic oscillator* potential $V = \frac{1}{2} m \omega^2 x^2$ with $\omega = 2a$, and energy eigenvalue $E = \hbar a = \frac{1}{2} \hbar \omega$ — exactly the *ground state* of the quantum harmonic oscillator.

(c) Expectation Values

— $\langle x \rangle$ and $\langle x^2 \rangle$

Since $|\Psi|^2 \sim e^{-2am\frac{x^2}{\hbar}}$ is an *even function*:

$$\boxed{\langle x \rangle = 0}$$

For $\langle x^2 \rangle$, use the Gaussian integral with $\alpha = 2a\frac{m}{\hbar}$:

$$\langle x^2 \rangle = A^2 \cdot \frac{1}{2\alpha} \sqrt{\frac{\pi}{\alpha}} = \sqrt{\frac{\alpha}{\pi}} \cdot \frac{1}{2\alpha} \sqrt{\frac{\pi}{\alpha}} = \frac{1}{2\alpha}$$

$$\boxed{\langle x^2 \rangle = \frac{\hbar}{4am}}$$

— $\langle p \rangle$ and $\langle p^2 \rangle$

For $\langle p \rangle$ at $t = 0$ (wave function is real):

$$\langle p \rangle = -i\hbar \int \Psi \frac{\partial \Psi}{\partial x} dx = -i\hbar \cdot \left(-2a\frac{m}{\hbar}\right) A^2 \int x e^{-\alpha x^2} dx = 0$$

The integral vanishes because the integrand is *odd*:

$$\boxed{\langle p \rangle = 0}$$

For $\langle p^2 \rangle$, use the result from part (b):

$$\begin{aligned} \langle p^2 \rangle &= -\hbar^2 \int \Psi^* \frac{\partial^2 \Psi}{\partial x^2} dx = -\hbar^2 \left(-\frac{2am}{\hbar} + \frac{4a^2 m^2}{\hbar^2} \langle x^2 \rangle \right) \\ &= 2am\hbar - 4a^2 m^2 \cdot \frac{\hbar}{4am} = 2am\hbar - am\hbar \end{aligned}$$

$$\boxed{\langle p^2 \rangle = am\hbar}$$

“**Cross-check via energy:** $\langle E \rangle = \hbar a = \langle p^2 \rangle \frac{1}{2m} + \langle V \rangle = \frac{am\hbar}{2m} + 2a^2 m \cdot \frac{\hbar}{4am} = \frac{a\hbar}{2} + \frac{a\hbar}{2} = a\hbar$ ”

(d) Uncertainties and the Heisenberg Principle

— Standard deviations

$$\sigma_x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{\hbar}{4am}} = \frac{1}{2} \sqrt{\frac{\hbar}{am}}$$

$$\sigma_p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \sqrt{am\hbar}$$

— Product

$$\sigma_x \cdot \sigma_p = \frac{1}{2} \sqrt{\frac{\hbar}{am}} \cdot \sqrt{am\hbar} = \frac{1}{2} \sqrt{\hbar^2}$$

$$\sigma_x \sigma_p = \frac{\hbar}{2}$$

— Consistency with the uncertainty principle

The Heisenberg uncertainty principle states:

$$\sigma_x \sigma_p \geq \frac{\hbar}{2}$$

Our result gives *exactly* $\frac{\hbar}{2}$ — the *minimum possible value*. This is consistent with and saturates the uncertainty principle.

“**Why equality?** The Gaussian wave packet is the *unique* wave function that minimizes the position-momentum uncertainty product. This wave function is precisely the ground state of the quantum harmonic oscillator — a well-known minimum-uncertainty state.”

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Problem 2

For a particle in the harmonic oscillator potential, its ground state is defined via the lowering operator

$$\hat{a}_- = \frac{1}{\sqrt{2\hbar m\omega}}(i\hat{p} + m\omega\hat{x})$$

by requiring $\hat{a}_- |0\rangle = 0$.

- (a) Find the normalized wave function of the ground state $\Psi_0(x) = \langle x | 0 \rangle$ in the position space.
- (b) For the first excited state $|1\rangle$, it can be obtained from $|1\rangle = \hat{a}_+ |0\rangle$, where the raising operator reads $\hat{a}_+ = \frac{1}{\sqrt{2\hbar m\omega}}(-i\hat{p} + m\omega\hat{x})$, find normalized $\langle x | 1 \rangle$.
- (c) Find the expectation values of the potential energy and the kinetic energy in the n -th stationary state.

“**Hints:** $\hat{a}_- |n\rangle = \sqrt{n} |n-1\rangle$ and $\hat{a}_+ |n\rangle = \sqrt{n+1} |n+1\rangle$.”

Solution.

(a) Ground State Wave Function $\Psi_0(x)$

— Setting up the condition

The ground state satisfies $\hat{a}_- |0\rangle = 0$. In position space with $\hat{p} = -i\hbar \frac{\partial}{\partial x}$:

$$\frac{1}{\sqrt{2\hbar m\omega}} \left(\hbar \frac{\partial}{\partial x} + m\omega x \right) \Psi_0(x) = 0$$

This gives the *first-order ODE*:

$$\frac{d\Psi_0}{dx} = -\frac{m\omega}{\hbar} x \Psi_0$$

— Solving by separation

$$\frac{d\Psi_0}{\Psi_0} = -\frac{m\omega}{\hbar} x dx \Rightarrow \Psi_0(x) = C \exp\left(-\frac{m\omega}{2\hbar} x^2\right)$$

— Normalization

$$|C|^2 \int_{-\infty}^{\infty} e^{-m\omega \frac{x^2}{\hbar}} dx = 1$$

Using $\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}$ with $\alpha = m\frac{\omega}{\hbar}$:

$$|C|^2 \sqrt{\frac{\pi \hbar}{m\omega}} = 1 \Rightarrow C = \left(\frac{m\omega}{\pi \hbar}\right)^{\frac{1}{4}}$$

$$\boxed{\Psi_0(x) = \left(\frac{m\omega}{\pi \hbar}\right)^{\frac{1}{4}} \exp\left(-\frac{m\omega x^2}{2\hbar}\right)}$$

(b) First Excited State $\Psi_1(x)$

— Apply the raising operator

$$|1\rangle = \hat{a}_+ |0\rangle \Rightarrow \Psi_1(x) = \frac{1}{\sqrt{2\hbar m\omega}} (-i\hat{p} + m\omega x) \Psi_0(x)$$

Since $\hat{p} = -i\hbar \frac{\partial}{\partial x}$:

$$-i\hat{p} = -\hbar \frac{\partial}{\partial x}$$

Computing the action on Ψ_0 :

$$-\hbar \frac{\partial \Psi_0}{\partial x} = -\hbar \left(-\frac{m\omega x}{\hbar}\right) \Psi_0 = m\omega x \Psi_0$$

Therefore:

$$\begin{aligned} \Psi_1(x) &= \frac{1}{\sqrt{2\hbar m\omega}} (m\omega x + m\omega x) \Psi_0(x) = \frac{2m\omega x}{\sqrt{2\hbar m\omega}} \Psi_0(x) \\ &= \sqrt{\frac{2m\omega}{\hbar}} x \Psi_0(x) \end{aligned}$$

$$\boxed{\Psi_1(x) = \left(\frac{m\omega}{\pi \hbar}\right)^{\frac{1}{4}} \sqrt{\frac{2m\omega}{\hbar}} x \exp\left(-\frac{m\omega x^2}{2\hbar}\right)}$$

This is an *odd function* times a Gaussian — the characteristic shape of the first excited state.

(c) Expectation Values of KE and PE in the n -th State

— Express operators via ladder operators

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}_+ + \hat{a}_-), \quad \hat{p} = i\sqrt{\frac{m\omega \hbar}{2}} (\hat{a}_+ - \hat{a}_-)$$

— Potential Energy: $\langle V \rangle$

$$\hat{x}^2 = \frac{\hbar}{2m\omega} (\hat{a}_+^2 + \hat{a}_+ \hat{a}_- + \hat{a}_- \hat{a}_+ + \hat{a}_-^2)$$

Taking the expectation value in state $|n\rangle$, *only the diagonal terms survive* ($\hat{a}_+^2$ and $\hat{a}_-^2$ change n by ± 2 , the cross terms change it by 0 but we need the diagonal part):

$$\langle n | \hat{x}^2 | n \rangle = \frac{\hbar}{2m\omega} (\langle n | \hat{a}_+ \hat{a}_- | n \rangle + \langle n | \hat{a}_- \hat{a}_+ | n \rangle)$$

Using the hints:

$$\hat{a}_+ \hat{a}_- |n\rangle = \sqrt{n} \hat{a}_+ |n-1\rangle = n |n\rangle$$

$$\hat{a}_- \hat{a}_+ |n\rangle = \sqrt{n+1} \hat{a}_- |n+1\rangle = (n+1) |n\rangle$$

Therefore:

$$\langle \hat{x}^2 \rangle_n = \frac{\hbar}{2m\omega} (2n+1)$$

$$\langle V \rangle_n = \frac{1}{2} m \omega^2 \langle x^2 \rangle_n = \frac{1}{4} (2n+1) \hbar \omega = \frac{1}{2} E_n$$

— Kinetic Energy: $\langle T \rangle$

$$\hat{p}^2 = -\frac{m\omega\hbar}{2}(\hat{a}_+^2 - \hat{a}_+\hat{a}_- - \hat{a}_-\hat{a}_+ + \hat{a}_-^2)$$

The diagonal terms give:

$$\langle \hat{p}^2 \rangle_n = -\frac{m\omega\hbar}{2}(-\langle \hat{a}_+\hat{a}_- \rangle_n - \langle \hat{a}_-\hat{a}_+ \rangle_n) = \frac{m\omega\hbar}{2}(2n+1)$$

$$\langle T \rangle_n = \langle p^2 \rangle_n = \frac{1}{4} (2n+1) \hbar \omega = \frac{1}{2} E_n$$

Summary

Quantity	Result	Fraction of E_n
Energy eigenvalue	$E_n = (n + \frac{1}{2})\hbar\omega$	—
$\langle V \rangle_n$	$\frac{1}{4}(2n+1)\hbar\omega$	$\frac{1}{2}$
$\langle T \rangle_n$	$\frac{1}{4}(2n+1)\hbar\omega$	$\frac{1}{2}$
$\langle V \rangle_n + \langle T \rangle_n$	$(n + \frac{1}{2})\hbar\omega = E_n$	1

“**Physical significance:** The result $\langle T \rangle_n = \langle V \rangle_n = \frac{E_n}{2}$ is the *quantum virial theorem* for the harmonic oscillator — kinetic and potential energies share the energy equally, independent of n .”

Problem 3

Orthonormal basis $|1\rangle, |2\rangle, |3\rangle$ with $\langle i | j \rangle = \delta_{ij}$.

$$|\alpha\rangle = i|1\rangle - 2|2\rangle - i|3\rangle, \quad |\beta\rangle = i|1\rangle + 2|3\rangle$$

- Construct $\langle\alpha|$ and $\langle\beta|$.
- Find inner products $\langle\alpha|\beta\rangle$ and $\langle\beta|\alpha\rangle$.
- Find the matrix elements of $\hat{A} = |\alpha\rangle\langle\beta|$. Is it Hermitian?

Solution.

(a) Construct $\langle\alpha|$ and $\langle\beta|$

To get the *bra*, we take the *complex conjugate of every coefficient* and write the dual basis in reverse order:

$$\langle\alpha| = -i\langle 1| - 2\langle 2| + i\langle 3|$$

$$\langle\beta| = -i\langle 1| + 2\langle 3|$$

“Note: $\langle\beta|$ has *zero* coefficient for $\langle 2|$ since $|\beta\rangle$ has no $|2\rangle$ component.”

(b) Inner Products $\langle\alpha|\beta\rangle$ and $\langle\beta|\alpha\rangle$

— $\langle\alpha|\beta\rangle$

$$\langle\alpha|\beta\rangle = (-i)(i) + (-2)(0) + (i)(2) = 1 + 0 + 2i$$

$$\langle\alpha|\beta\rangle = 1 + 2i$$

— $\langle\beta|\alpha\rangle$

$$\langle \beta | \alpha \rangle = (-i)(i) + (0)(-2) + (2)(-i) = 1 + 0 - 2i$$

$$\boxed{\langle \beta | \alpha \rangle = 1 - 2i}$$

— Verification of the conjugate-symmetry property

$$(\langle \beta | \alpha \rangle)^* = (1 - 2i)^* = 1 + 2i = \langle \alpha | \beta \rangle$$

$$\boxed{\langle \alpha | \beta \rangle = \langle \beta | \alpha \rangle^*}$$

This confirms the fundamental property of the inner product in a complex vector space.

(c) Matrix Elements of $\hat{A} = |\alpha\rangle\langle\beta|$

The matrix elements are $A_{ij} = \langle i | \hat{A} | j \rangle = \langle i | \alpha \rangle \langle \beta | j \rangle = \alpha_i \beta_j^*$.

— Building the coefficient vectors

	$ 1\rangle$	$ 2\rangle$	$ 3\rangle$
$ \alpha\rangle: \alpha_i$	i	-2	$-i$
$\langle\beta : \beta_j^*$	$-i$	0	2

— Computing all 9 elements

$$A_{ij} = \alpha_i \cdot \beta_j^*$$

	$j = 1 \ (\beta_1^* = -i)$	$j = 2 \ (\beta_2^* = 0)$	$j = 3 \ (\beta_3^* = 2)$
$i = 1 \ (\alpha_1 = i)$	$(i)(-i) = 1$	0	$(i)(2) = 2i$
$i = 2 \ (\alpha_2 = -2)$	$(-2)(-i) = 2i$	0	$(-2)(2) = -4$
$i = 3 \ (\alpha_3 = -i)$	$(-i)(-i) = -1$	0	$(-i)(2) = -2i$

— The matrix

$$\boxed{\mathbf{A} = \begin{bmatrix} 1 & 0 & 2i \\ 2i & 0 & -4 \\ -1 & 0 & -2i \end{bmatrix}}$$

— Is it Hermitian?

Check if $A^\dagger = A$, i.e., $A_{ji}^* = A_{ij}$:

Element	A_{ij}	A_{ji}^*	Equal?
(1, 3)	$2i$	$(-1)^* = -1$	No
(3, 1)	-1	$(2i)^* = -2i$	No

Since $A_{13} = 2i \neq -1 = A_{31}^*$:

$$\boxed{\hat{A} = |\alpha\rangle\langle\beta| \text{ is NOT Hermitian.}}$$

“**General rule:** $|\alpha\rangle\langle\beta|$ is Hermitian *only if* $|\alpha\rangle = e^{i\varphi} |\beta\rangle$ for some phase φ — which is clearly not the case here since $|\alpha\rangle$ and $|\beta\rangle$ differ in both the $|2\rangle$ component and the relative phases of their coefficients.”

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Problem 4

A three-level quantum system has Hamiltonian and observables:

$$H = \hbar\omega \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad A = \lambda \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad B = \mu \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 3 \\ 0 & 3 & 0 \end{bmatrix}$$

- (a) Find the eigenvalues and normalized eigenvectors of H , A , and B .
- (b) At $t = 0$, the system is in state $|S(0)\rangle = (c_1, c_2, c_3)^T$ with $|c_1|^2 + |c_2|^2 + |c_3|^2 = 1$. Find $\langle H \rangle$, $\langle A \rangle$, and $\langle B \rangle$ at $t = 0$.
- (c) Find $|S(t)\rangle$ at arbitrary time t . If energy is measured, what are the possible outcomes and their probabilities?

Solution.

(a) Eigenvalues and Eigenvectors

— Hamiltonian H

$$H = \hbar\omega \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Already diagonal! The eigenvalues are read directly from the diagonal:

$$E_1 = \hbar\omega, \quad E_2 = \hbar\omega, \quad E_3 = 2\hbar\omega$$

Eigenvectors (the standard basis):

$$|1\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad |2\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad |3\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

“Note: $E_1 = E_2 = \hbar\omega$ is a *degenerate* eigenvalue. Any linear combination of $|1\rangle$ and $|2\rangle$ is also an eigenvector with eigenvalue $\hbar\omega$.”

— Observable A

$$A = \lambda \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

The matrix is *block-diagonal*: a 2×2 block (rows 1—2) and a 1×1 block (row 3).

Block 1 — the 2×2 submatrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$:

$$\det \begin{bmatrix} -\nu & 1 \\ 1 & -\nu \end{bmatrix} = \nu^2 - 1 = 0 \Rightarrow \nu = \pm 1$$

Block 2 — the 1×1 submatrix (2): eigenvalue $\nu = 2$.

Eigenvalue	Eigenvector
$-\lambda$	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$
$+\lambda$	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$
2λ	$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

— Observable B

$$B = \mu \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 3 \\ 0 & 3 & 0 \end{bmatrix}$$

Again block-diagonal: a 1×1 block (row 1) and a 2×2 block (rows 2—3).

Block 1 (2): eigenvalue 2μ .

Block 2 $\begin{bmatrix} 0 & 3 \\ 3 & 0 \end{bmatrix}$:

$$\det \begin{bmatrix} -\nu & 3 \\ 3 & -\nu \end{bmatrix} = \nu^2 - 9 = 0 \Rightarrow \nu = \pm 3$$

Eigenvalue	Eigenvector
2μ	$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
-3μ	$\frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$
$+3\mu$	$\frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

(b) Expectation Values at $t = 0$

Given $|S(0)\rangle = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$ with $|c_1|^2 + |c_2|^2 + |c_3|^2 = 1$.

— $\langle H \rangle$

Since H is diagonal:

$$\langle H \rangle = |c_1|^2(\hbar\omega) + |c_2|^2(\hbar\omega) + |c_3|^2(2\hbar\omega)$$

$$\langle H \rangle = (|c_1|^2 + |c_2|^2 + 2|c_3|^2) \hbar\omega = (1 + |c_3|^2) \hbar\omega$$

— $\langle A \rangle$

Computing $\langle A \rangle = \sum_{i,j} c_i^* A_{ij} c_j$:

$$= \lambda(c_1^* c_2 + c_2^* c_1 + 2|c_3|^2)$$

$$\langle A \rangle = \lambda(2 \operatorname{Re}(c_1^* c_2) + 2|c_3|^2)$$

— $\langle B \rangle$

$$\langle B \rangle = \mu(2|c_1|^2 + 3c_2^* c_3 + 3c_3^* c_2)$$

$$\langle B \rangle = \mu(2|c_1|^2 + 6 \operatorname{Re}(c_2^* c_3))$$

(c) Time Evolution and Energy Measurements

— Finding $|S(t)\rangle$

Since H is diagonal in the given basis, each component evolves with its own phase:

$$e^{-iH\frac{t}{\hbar}} = \begin{bmatrix} e^{-i\omega t} & 0 & 0 \\ 0 & e^{-i\omega t} & 0 \\ 0 & 0 & e^{-2i\omega t} \end{bmatrix}$$

$$|S(t)\rangle = \begin{bmatrix} c_1 e^{-i\omega t} \\ c_2 e^{-i\omega t} \\ c_3 e^{-2i\omega t} \end{bmatrix}$$

— Energy measurement outcomes

The possible energy eigenvalues are $E = \hbar\omega$ and $E = 2\hbar\omega$.

For $E = \hbar\omega$: This eigenvalue is *degenerate* (eigenspace spanned by $|1\rangle$ and $|2\rangle$). The total probability is:

$$P(E = \hbar\omega) = |c_1|^2 + |c_2|^2$$

For $E = 2\hbar\omega$: Non-degenerate (eigenspace spanned by $|3\rangle$):

$$P(E = 2\hbar\omega) = |c_3|^2$$

— Summary

$E = \hbar\omega$ probability: $ c_1 ^2 + c_2 ^2$ $E = 2\hbar\omega$ probability: $ c_3 ^2$

“**Check:** $(|c_1|^2 + |c_2|^2) + |c_3|^2 = 1$ ”

“**Key observation:** The degenerate eigenvalue $\hbar\omega$ means that the system “hides” which state ($|1\rangle$ or $|2\rangle$) contributed — we can only measure that the energy is $\hbar\omega$, not which degenerate eigenvector was responsible.”

